

Session 5: Data Selection through FineWeb

Turning “Quality” into a Testable Pipeline

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Training Language Models: From First Principles to Efficient Serving

Lecture map

- 1 Web data becomes a training distribution
- 2 Deduplication and learned selection
- 3 A fair data ablation

Guiding question

How can a filtering decision be evaluated before and after spending compute on pretraining?

What should be clear by the end?

- 1 Reconstruct the stages and design logic of the FineWeb pipeline.
- 2 Distinguish extraction, filtering, deduplication, and learned selection.
- 3 Explain why plausible heuristics still require ablations.
- 4 Design a fair token-matched comparison with contamination controls.

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Data quality is an operational claim

A useful definition

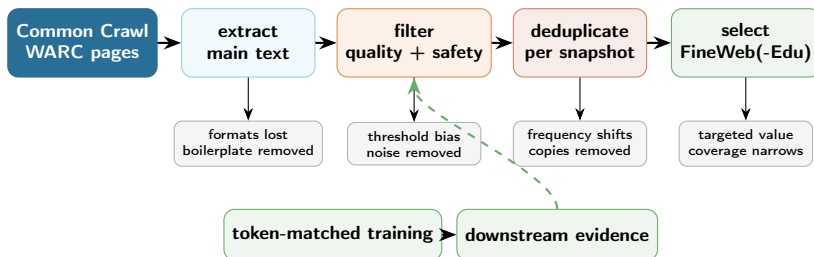
Corpus A is better than corpus B for a stated goal if training under the same compute, token, model, and evaluation contract yields stronger evidence on that goal.

What “cleaner” does not establish

Human preference, lower toxicity, fewer duplicates, and benchmark performance are related but different objectives. Name which one the selector targets.

Selection changes the model's empirical world. It is part of the model design.

FineWeb turns Common Crawl into a staged experiment



The recipe is validated by the model it produces, not by cleanliness alone.

Source: Penedo et al., The FineWeb Datasets and Hugging Face FineWeb blog

Extraction is already a selection function

HTML contains navigation, cookie dialogs, comments, ads, hidden text, and main prose.

High recall extractor

Keeps more useful prose but also more boilerplate and malformed text.

High precision extractor

Produces cleaner documents but may erase code, lists, tables, and minority formats.

Inspect raw HTML, extracted text, and rejection reason together. Otherwise, the selection mechanism is invisible.

Base filters encode hypotheses about usefulness

Signal	Hypothesis	Risk
character or word length	fragments teach little	removes concise answers
repeated lines or n-grams	boilerplate is low value	removes poetry and code
symbol ratios	noise has unusual composition	harms math-heavy pages
language score	target language dominates	harms multilingual content
blocked URLs or domains	known sources are undesirable	domain bias

Every threshold is an ablation candidate

Count removals, sample both sides of the threshold, and compare downstream behavior.

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Duplicate is not one relation

Exact

identical normalized text
or hash

Near duplicate

high overlap of word or
character shingles

Semantic

same information with
different surface form

Design choices

Normalization, shingle size, similarity threshold, representative choice, and deduplication scope all change what survives.

MinHash estimates Jaccard similarity cheaply

For shingle sets A and B :

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}$$

A random min-wise hash satisfies:

$$\Pr[\min h(A) = \min h(B)] = J(A, B)$$

Repeating hashes creates a signature. Locality-sensitive buckets generate candidate pairs without comparing every document pair.

Interpretation

MinHash approximates set overlap. It is not a semantic similarity oracle.

FineWeb made deduplication scope an empirical question

Global deduplication

Removes copies across all crawl snapshots. It maximizes novelty but can strongly alter the time and source distribution.

Per-snapshot deduplication

Removes repeated pages inside a crawl while preserving recurrence across time.

Lesson from the reported ablation

More aggressive deduplication did not automatically yield better downstream models. A sensible mechanism still needs token-matched training evidence.

FineWeb-Edu learns a narrower notion of value

- 1 An LLM scores pages for educational usefulness.
- 2 A smaller classifier learns to imitate those labels at corpus scale.
- 3 Thresholds trade retained volume against the targeted quality signal.

Potential benefit

Scalable selection for explanatory, knowledge-rich text.

Potential bias

The teacher rubric and classifier can narrow style, language, and topic coverage.

Checkpoint: selector disagreement

Think, then compare

A heuristic rejects a document that the educational classifier scores highly. What evidence would help decide whether the conflict reveals noise, a blind spot, or a genuinely different objective?

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Match the experiment, not only the dataset size

Keep constant:

- model initialization and architecture
- tokenizer and sequence length
- tokens seen, optimizer updates, and learning-rate schedule
- evaluation token budget and benchmark harness
- hardware and throughput measurement method

Replacement policy matters

If filtering removes data, decide whether to train on fewer unique tokens, resample survivors, or draw more raw data. These test different claims.

Evaluate the selector before training

Retention profile

tokens and documents kept
language and domain shifts
length and source distributions

Error analysis

sample accepted and rejected pages
label false positives and false
negatives
inspect threshold neighborhoods

A selector can be precise on average and still erase a small but important domain.

Evaluate the trained models at equal budgets

- 1 Compare validation loss on declared distributions.
- 2 Measure downstream tasks relevant to the selection objective.
- 3 Report training throughput and tokens required to reach a target loss.
- 4 Repeat enough seeds to separate a trend from run noise.

A strong claim

“At equal tokens and architecture, filter F reaches the same validation loss in fewer updates on distribution D .”

Contamination can imitate data quality

- Search for benchmark examples and close variants before training.
- Split related pages and document versions together.
- Record the order of deduplication, split assignment, and benchmark filtering.
- Report contaminated and cleaned evaluations separately when needed.

Core warning

Higher benchmark accuracy is not evidence of better generalization if evaluation text leaked into pretraining.

Checkpoint: pre-register the comparison

Think, then compare

Write one target claim, the baseline corpus, the selected corpus, the token replacement policy, two pre-training diagnostics, and two post-training metrics.

The durable ideas

- 1 Extraction and filtering define the training distribution before tokenization begins.
- 2 Deduplication has a similarity definition, threshold, representative rule, and scope.
- 3 FineWeb shows that plausible pipeline choices need controlled training ablations.
- 4 Contamination controls and token-matched budgets are prerequisites for quality claims.